# Molecular Biology and Genetics

## Science

**May, 2026**

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

# Molecular Biology and Genetics (A11365)

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| **Short Title:** | Molecular Biology and Genetics | | |
| **Faculty** | Science and Computing | | |
| **Department** | Science | | |
| **Cluster** | Biology | | |
| **Author** | AHEARNE | | |
| **Credits** | 5 | **Level:** | Intermediate |

## Description of Module / Aims

This module builds on previous modules such as 'Scientific Enquiry in Biology' and 'Microbial Biotechnology & Genetics' to further develop core concepts in cell biology, genetics and molecular biology.

## Programmes

| | | |
|---|---|---|
| `SCIE-0058` | BSc (Hons) in Molecular Biology with Biopharmaceutical Science (`WD_SMBIO_B`) | 3 / 5 / M |
| `SCIE-0058` | BSc in Molecular Biology with Biopharmaceutical Science (`WD_SMBIO_D`) | 3 / 5 / M |

## Indicative Content

- DNA structure
- The molecular biology of transcription and its regulation: RNA polymerases, promoter sequences, regulatory elements, transcription factors
- RNA processing and its control: capping, polyadenylation, intron splicing, sequence editing; miRNA
- The molecular biology of translation and its regulation: protein synthesis by the ribosome
- Gene and genome structure analysis: preparation of cloned DNA, restriction mapping, Southern blotting, location of transcriptional start point, chromosomal location and DNA sequencing
- Gene cloning: PCR, vectors and their uses, enzymes used for manipulating DNA, construction and uses of recombinant DNA, transformation and screening
- New developments in molecular biology will be introduced as appropriate
- Introduction to bioinformatics

## Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Appraise the stages of gene expression: DNA transcription, RNA processing, mRNA translation and post-translational modifications of proteins.
2. Interpret the regulation of the processes of gene expression by a variety of mechanisms.
3. Relate the overall structure and organisation of typical prokaryotic and eukaryotic genomes to gene expression and regulation.
4. Analyse the practical approaches used in gene and genome analysis.
5. Use laboratory techniques to design simple strategies for gene cloning, manipulation of DNA and characterisation of genes.
6. Communicate results of laboratory work using a combination of formal written laboratory reports and worksheets.

## Learning and Teaching Methods

- Lectures.
- Lab-based practicals.

- Self-study.

## Assessment Methods

| | Weighing | Outcomes Assessed |
|---|---|---|
| Final Written Examination | 60% | 1, 2, 3, 4 |
| Continuous Assessment | 40% | |
| *Practical* | 40% | 5,6 |

## Assessment Criteria

**<40%:** Little or no demonstration of an understanding of the module content. Lack of competence with associated laboratory skills & practices & an inability to carry out critical thinking.

**40%-49%:** Will show a limited understanding of the module content covered. May have some difficulty with applying critical thinking and problem-solving. Will display basic ability with associated laboratory skills & practices.

**50%-59%:** As well as the above, the student will show a good understanding of the module content with demonstration of some critical thinking & problem-solving skills. Will display good competence with associated laboratory skills & practices.

**60%-69%:** In addition to the above, will demonstrate more detailed knowledge of the module content, very good critical thinking skills & a high level of competence & efficiency in associated laboratory practicals.

**70%-100%:** All of the above to an excellent level.

**Note:**

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

## Learning Modes

| Learning Mode | F/T Hours | P/T Hours |
|---|---|---|
| Lecture | 24 | |
| Practical | 24 | |
| Independent Learning | 87 | |

## Essential Material

- Alberts, B., D. Bray, K. Hopkin, A. Johnson, J. Lewis and Raff, M., Roberts, K. & Walter, P. _Essential Cell Biology_. 3rd ed. NY, USA & Abingdon, UK: Garland Science, 2010.
- Madigan, M.T., J.M. Martinko, D.A. Stahl, D. Clark, K.S. Bender and Buckley, D.H. _Brock's Biology of Micro-organisms_. 13th ed. Upper Saddle River, NJ: Prentice Hall / Pearson Education, 2011.

## Requested Resources

- Room Type: Biology Lab